

Work and Practical Session: Machine Learning

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Session 03:

Supervised Learning – Decision Trees

Exercise #1: Decision Trees

1. Let X be the random variable representing an event. Suppose that the probabilities attached to the 5 outcomes o_i of the event are $P(X=o_1)=0,5$ and $P(X=o_2) = P(X=o_3) = P(X=o_4) = P(X=o_5) = 0,125$.
 E_1 is the entropy of the random variable X . Compute E_1 . Write down in words what it means.
2. Consider the entropy E_2 of a random variable where the probabilities of the 5 outcomes are equiprobable. Without computing it, is E_2 lower or greater than E_1 ?
3. Make a decision tree and a Perceptron network that compute the logical AND function. How does the decision tree compare to the Perceptron solution?
4. Apply the ID3 algorithm manually to turn this politically incorrect data from Quinlan into a decision tree in order to classify which attributes make a person attractive, and then extract the rules.

Height	Hair	Eyes	Attractive?
Small	Blonde	Brown	No
Tall	Dark	Brown	No
Tall	Blonde	Blue	Yes
Tall	Dark	Blue	No
Small	Dark	Blue	No
Tall	Red	Blue	Yes
Tall	Blonde	Brown	No
Small	Blonde	Blue	Yes